

SUBHRAJIT DAS

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EDUCATION

PhD in Computer Science and Engineering

2025 - Present

Indian Institute of Technology Gandhinagar, Gandhinagar, India

Director's PhD Fellow

Advisors: Prof. Abhishek Bichhawat, Prof. Yuvraj Patel

CPI: 10

M.Tech. in Computer Science and Engineering

2023 - 2025

Indian Institute of Technology Gandhinagar, Gandhinagar, India

CPI: 9.84

Specialization: Computer Systems

M.Sc. in Computer Science

2021 - 2023

University of Kalyani, Kalyani, India

CGPA: 9.76

First Class with Distinction

First Rank in University

B.Sc. (Honours) in Computer Science

2018 - 2021

Panihati Mahavidyalaya, Sodepur, India (*Affiliated to WBSU*)

CGPA: 9.89

RESEARCH INTERESTS

CPU Parallelism, Distributed Systems, and Usable Security

RESEARCH EXPERIENCE

Accelerating Large Integer Arithmetic with Parallel Addition, Subtraction, and Vedic-Based Multiplication Using AVX512 [2]

Jan 2024 - Apr 2025

Supervisors: Prof. Abhishek Bichhawat, Prof. Yuvraj Patel

M.Tech Thesis, IIT Gandhinagar

- Designed high-performance faster data-parallel algorithms for large integer addition and subtraction using AVX512, achieving average execution-time speedups of $2\times$ against the GNU Multiple-Precision Arithmetic Library (GMP).

Studies on Various Maximal Covering Location Problems using Genetic and Artificial Bee Colony Algorithms

Sep 2022 - Jun 2023

Supervisors: Prof. Priya Ranjan Sinha Mahapatra and Dr. Soumen Atta

M.Sc Thesis, University of Kalyani

- Designed algorithms to solve the NP-hard probabilistic maximal covering location allocation problem using artificial bee colony algorithm, achieving optimal benchmark results of commercial solver CPLEX in 50% of cases, matching accuracy with other meta-heuristics models while beating most of them in computational time.

Reversible Multiplier Accumulate Unit

Jan 2021 - Aug 2021

Supervisors: Mr. Biswanath Sen

B.Sc Project, Panihati Mahavidyalaya

- Proposed a reversible design of the multiplier accumulate unit (MAC) using reversible gates for low power consumption and heat dissipation, helping us in energy saving. Additionally, proposed a reversible design of adder/subtractor and information shifter for low power consumption.

TEACHING EXPERIENCE

Teaching Assistant

Dept. of CSE

Jul 2023 - Present

IIT Gandhinagar

- Assisted with courses such as Distributed Systems and Cloud Computing, Computer & Network Security, Compilers, and Data Structures and Algorithms - I.

Guest Lecture on SIMD and Code Profiling

Computer Systems, CS 612

Oct, 2025

IIT Gandhinagar

- Given a couple of guest lectures on exploiting x86-64 data parallelism using compiler auto-vectorization, intrinsics and also profiling code with Linux perf and relevant syscalls for the course CS 612 - Computer Systems.

Guest Lecture on RAID

Operating Systems, CS 330

April, 2025

IIT Gandhinagar

- Given a guest lecture on RAID for the course CS 330 - Operating Systems.

Principal Instructor

Dept. of CSE

Nov 2024

IIT Gandhinagar

- Conducted a student-run course (SRC) titled “Code Profiling and Optimization” in collaboration with a colleague, under the mentorship of *Prof. Abhishek Bichhawat*. The course focused on code profiling, performance benchmarking, and leveraging various tools and libraries for code optimization.

PROJECTS

Distributing GEMM over Network of Stations [3]

Team Contributor

Oct 2025 - Present

IIT Gandhinagar & University of Edinburgh

- Investigating how a cluster of commodity CPUs together can help for massive data-parallel compute-intensive task like general matrix multiplication (GEMM) which is typically reserved for GPUs.

Developer Experience of Consent Management Platform Integration

Team Contributor

Oct 2025 - Present

IIT Gandhinagar

- Investigating the usability challenges developers face when integrating consent management platforms (CMPs) into web applications. Manuscript under revision for submission to top-tier security venues.

Online Authentication Habits of Indian Users [1]

Team Contributor

Oct 2023 - May 2024

IIT Gandhinagar

- Conducted a structured survey with 90 Indian participants, analyzing awareness, usage, and perceptions regarding password habits, password managers, and two-factor authentication (2FA). Highlighted many interesting insights, including a tendency to use default settings, and emphasized the need for tailored strategies to enhance password security.

Instant Payment Gateway

Team Contributor

Feb 2024 - April 2024

IIT Gandhinagar

- Developed an instant payment system using microservices architecture (Go, gRPC, Docker) with a single-server deployment, ensuring secure, fault-tolerant transactions via Nginx load balancing and sharded MySQL/SQLite. Designed key components (Authenticator, Payment Handler, Resolver, Banks) for transaction coordination, failure recovery, and notifications, leveraging ELK Stack and wrk for performance benchmarking.

ACCEPTED PUBLICATIONS

- [1] Pratyush Choudhary*, Subhrajit Das*, Mukul Paras Potta*, Prasuj Das, and Abhishek Bichhawat, “Online Authentication Habits of Indian Users,” in *2024 Conference on Building a Secure & Empowered Cyberspace (BuildSEC)*, 2024, pp. 66–73. DOI: 10.1109/BuildSEC64048.2024.00018.

PREPRINTS & ONGOING WORKS

- [2] Subhrajit Das, Abhishek Bichhawat, and Yuvraj Patel, “DigitsOnTurbo: Leveraging SIMD for Accelerating Large-number Arithmetic,” in *ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS) 2027*, **Under review**, 2027. arXiv: 2604.21566 [cs.DC].
- [3] Subhrajit Das*, Rachel Somerset*, Abhishek Bichhawat, and Yuvraj Patel, “Davids vs Goliath: Achieving GPU-like performance using Network of Stations,” in *European Conference on Computer Systems (EuroSys) 2027*, **Manuscript in preparation**, 2027.

POSITIONS OF RESPONSIBILITY

Volunteer

FSTTCS 2024

Dec 16-18, 2024

IIT Gandhinagar

- Volunteered at the 44th conference on Foundations of Software Technology and Theoretical Computer Science (FSTTCS 2024), organized by IARCS in association with ACM India.

Class Representative

Dept. of CSE, MCS

Oct 2021 - June 2023

University of Kalyani

- Facilitated communication between faculty, administration, and batch-mates in matters ranging from class-related concerns to advocating for a fee reduction for the batch.

ACHIEVEMENTS

Selected for Director’s Fellowship for admission into PhD at IIT Gandhinagar

Jan 2025

Best Paper Award for “Online Authentication Habits of Indian Users” at BuildSEC’24

Dec 2024

MoE Scholarship for Teaching Assistantship, IIT Gandhinagar

Jul 2023

All India Rank of 530 in the GATE 2023 Computer Science examination

Mar 2023

Awarded with UGC NET JRF in Computer Science

Dec 2022

Qualified for UGC NET in Computer Science for Lecturership

Dec 2022

SKILLS

Programming Languages

C, C++, Java, MATLAB, Assembly, Go, Python, R, Shell, SQL, Prolog, Kotlin, HTML, CSS, JavaScript, JSP

Tools/Libraries/Simulators

Git, Docker, gRPC, OpenMP, MPI, SIMD Intrinsics, Valgrind, Linux Perf, GDB, Wireshark, NS2, Mininet

Operating Systems

Linux/Unix, Windows, macOS